

Research Interests

I have academic and research experience in solving urban and regional planning problems through the application of geospatial and statistical tools, technologies, and policy-driven interventions. I am highly motivated to work on geospatial problems using machine learning, AI, and technology-driven methods, and I am eager to explore new technologies in this field.

Education

2026 - Expected Jan 2027	Masters in Urban & Regional Planning , Faculty of Civil Engineering, Khulna University of Engineering & Technology (KUET)
2020 - 2025	Bachelor in Urban & Regional Planning , Faculty of Civil Engineering, Khulna University of Engineering & Technology (KUET) CGPA- 3.97 on 4 (All semesters). Position in Class- 1 st in all academic years (2020-2025).
2017 - 2019	Higher Secondary Certificate (Science) , RAJUK Uttara Model College, GPA - 5.00 on 5.00

Thesis

Bachelor's Thesis

Title	Urban Growth Dynamics and Socioeconomic Drivers in Khulna: A Spatiotemporal Perspective
Supervisor	Professor Dr. Md. Manjur Morshed
Description	An urban expansion analysis using Remote Sensing and Socioeconomic Indicators (2000-2024) integrating binary classification, spatial statistics geographic distribution tools including JavaScript and geographic information systems.

Research & Publications (published as "Mst. Tasnia Nowrin")

- 1 **Al Kafy, A., Nowrin, M. T.,** Altuwaijri, H. A., & Mukarram, M. M. T. (2025). Advanced hybrid ensemble learning models for atmospheric humidity prediction: An environmental monitoring solution in arid regions. *International Journal of Environmental Science and Technology*. DOI: <https://doi.org/10.1007/s13762-025-06926-z>
- 2 **Nowrin, M. T.,** Rahman, M., Hasan, M. M., & Hasan, M. M. (2026). Price Hikes, Sustainability, and SDG Achievement Among Lower-Income Groups in Urban Bangladesh. Manuscript previously submitted; under revision for resubmission.
- 3 **Nowrin, M. T.** & Mohiuddin. T. (2025, February). Sustainable Desalination Technologies: A Solution to Coastal Water Scarcity in Bangladesh (Poster presentation). Plannation 3.0, Khulna University of Engineering & Technology (KUET), Khulna, Bangladesh.
- 4 Fattah, M. A., Rahman, M., Hasan, M. M., Hasan, M. M., **Nowrin, M. T.,** Morshed, S. R., Chakraborty, T., Al Kafy, A., Alsulamy, S., Khedher, K. M., & Shohan, A. A. A. (2023). Sustainability amidst economic strain: A quantitative exploration of COVID-19 induced food insecurity, obesity, and inflation impacts on health and sanitation in urban slums. (Unpublished manuscript)

Professional Experience

Oct 2024	Intern , Tiller , Dhaka, Bangladesh - Contributed to the "Preparation of Risk Sensitive Database for Core Area of Rangpur and Sylhet District Town (SRDP)" project. - Assisted in preparing land-use zoning and Mouza maps scaling using GIS for risk-sensitive planning. - Gained hands-on experience with urban spatial data processing and risk-sensitive zoning practices.
2023 -	Research Assistant , Center for Environmental Research (CER)

- Contributed to multidisciplinary research on environmental change, air pollution, transportation, and public health using GIS and statistical analysis (GEE, ArcGIS, Python).
- Assisted in drafting manuscripts, data collection and processing, and climate and transport-related modeling to provide policy-relevant, evidence-based solutions for a sustainable future.

Projects

- **Transportation Surveys**
Traffic surveys, AutoCAD, GIS and Excel for intersection and mobility analysis.
- **Regional Economics Analysis of Rajshahi Division**
Used PCA, SPSS, and GIS to map economic disparities.
- **Comparative Analysis of Smart Village Potential: Evaluating the Smartness of Botiaghata and Betaga Union**
Field and questionnaire surveys, GIS and SPSS-based of digital infrastructure in rural Bangladesh.
- **Assessment of Child Friendliness in Ispahani Colony, Khulna**
Conducted statistical analysis, field and questionnaire surveys.
- **Water Scarcity in Rajshahi Division: A feasibility Study for Sustainable Solutions**
Meta-analysis and GIS-based rainwater harvesting and recharge strategy.
- **Heritage Restoration – The Mosque City of Bagerhat**
Developed DPP & ToR with feasibility, budgeting, impact analysis & stakeholder mapping.
- **Sustainable Desalination Technologies: A Solution to Coastal Water Scarcity in Bangladesh**
PV-RO technology planning with GIS mapping for coastal water security.

Skills

Coding	Python, JavaScript, C, R
Tools & Softwares	ArcGIS, Google Earth Engine (GEE), Google Earth Pro, SPSS, Excel, AutoCAD, SketchUp, MS Office
Techniques	Network Analysis, Remote Sensing, GIS-Based Modeling, Spatial Analysis, Machine Learning
Professional	Project Management, Stakeholder Engagement, Teamwork, Leadership, Communication under Pressure
Language & Proficiency	Bangla (native), English (proficient); undergraduate program conducted in English (KUET, WHED-listed). IELTS planned.

Awards

2019–20	Dean's Award for Academic Excellence , Faculty of Civil Engineering, Khulna University of
2020–21	<i>Engineering & Technology (KUET)</i>
2021–22	Technical and Compensatory Scholarship , Khulna University of Engineering & Technology (KUET)

Leadership & Extracurricular

2025	Volunteer , Plannation 3.0, Khulna University of Engineering & Technology (KUET) Logistical Support, Event planning and team coordination.
2020 – 2024	Team Leader , Academic Lab Projects Led multiple projects in GIS and urban planning as well as contributed to surveying, research documentation, including literature reviews, technical reports, and academic presentations.
2020 – 2023	Co-Organizational Secretary , KUET Theatre Event planning and team coordination.